DNA Methylation Analysis: Statistics and Machine Learning in Breast Cancer

Thesis Work Presentation

Master's Degree in Mathematical Engineering

Elisabetta Roviera

November 24, 2025

Table of Contents

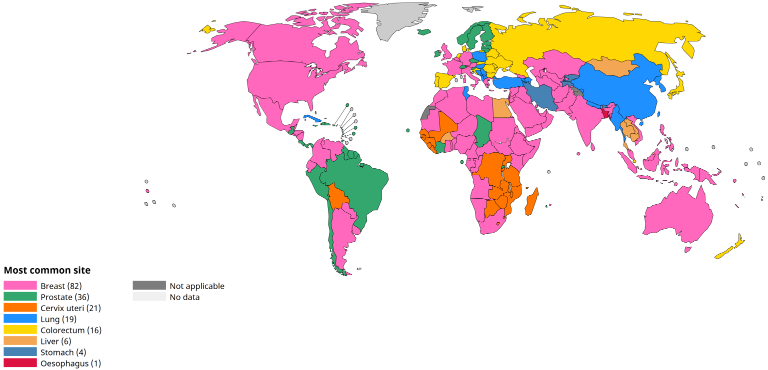
Introduction: Goals of the Presentation

- Background in **Mathematical Engineering** with a focus on **Data Analysis**.
- Goal: present a concrete thesis project integrating mathematics, computer science, and biomedicine.
- Case study: *DNA methylation* analysis in **breast cancer**.

A quantitative perspective on a real biomedical dataset.

Breast Cancer: Context and Relevance

- **High global incidence**, affecting millions of individuals.
- **Early detection** strongly improves patient outcomes.
- Why data matters: quantitative analysis supports **screening**, **diagnosis**, and **risk assessment**.

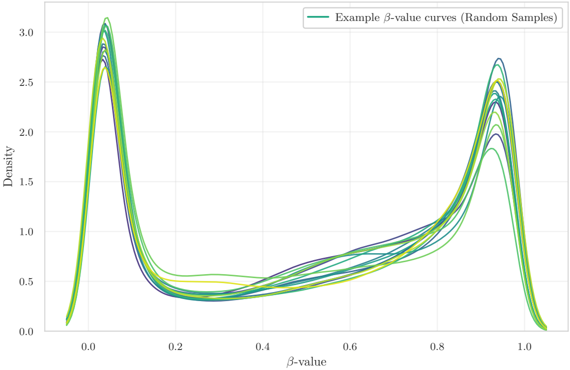


Most common cancer site per country, 2022 (excl. NMSC).

Source: Cancer TODAY — IARC, International Agency for Research on Cancer Globocan 2022.

CpG and β-values: The Language of Epigenetic Data

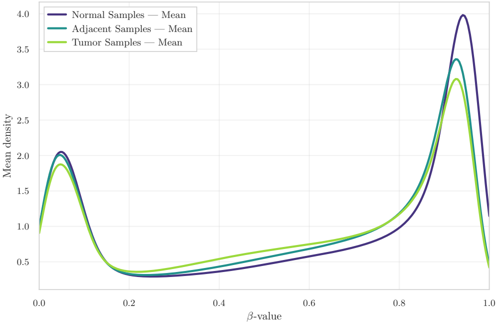
- **CpG sites**: genomic positions where DNA methylation can occur.
- **β-value** $\in [0, 1]$: proportion of methylation at each CpG.
- **DNA methylation** modulates **gene regulation**.
- In cancer, both **hypermethylation** and **hypomethylation** emerge.



Example β-value density curves from random samples. Internal dataset visualisation (thesis exploratory analysis).

Field Cancerization: The Central Research Question

- **Central question: is "normal" tissue near the tumor already epigenetically altered?**
- As Dr. Gambino highlights:
  "Identify early methylation alterations in normal tissues and assess their role in tumor initiation."
- Concept of **field cancerization**.



Mean β-value densities for Normal, Adjacent, and Tumor samples. Internal dataset visualisation (exploratory analysis).

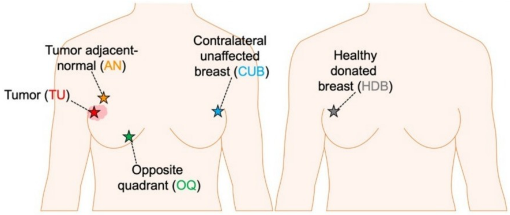


Illustration of sampling locations in breast tissue. Conceptual illustration of tumor, adjacent, and healthy breast tissue.

Overall Research Workflow

- **Consultation of medical, statistical, and computational literature** to understand biological context and methodological constraints.
- **Reproduction of key results from published studies** to confirm the consistency and reliability of each dataset.
- **Progressive methodological innovation**: integrating statistical filtering, stratification strategies, dimensionality reduction, and ultimately ML/DL modelling with careful hyperparameter tuning.

Goal: build a robust, reproducible, and biologically-consistent analysis pipeline.

Table of Contents

2 Data & Computational Challenges

Accessing and Understanding the Data

- GEO datasets: **heterogeneous formats** and **incomplete metadata**.
- Small n, huge p (up to $\sim$800k CpG sites).
- Raw **IDAT files** vs. preprocessed β**-value matrices**.
- **Unknown or inconsistent preprocessing** in some datasets.

Dataset	Samples (n)	CpGs (p)
GSE69914	397	485,512
GSE225845	477	750,426
GSE287331	311	702,573

Summary of the three GEO datasets used in the analysis.

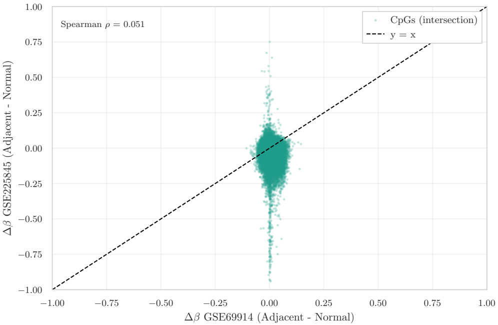
Computational Challenges

- **Huge files** $\rightarrow$ use of **Parquet** format, **LZ4** compression, chunked loading.
- Strict RAM constraints on Kaggle environments.
- Need for **Illumina manifest alignment** to correctly map probe IDs.

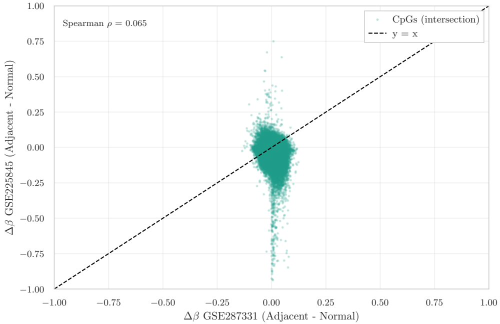
Memory-aware processing pipeline. From raw huge files to compressed Parquet storage, enabling analysis under strict RAM limits.

Can I Merge the Datasets? NO

- Different array platforms (**HM450** vs. **EPIC**).
- **Heterogeneous** and partly **unknown preprocessing pipelines**.
- Direct merge would introduce strong batch and platform effects.
- Decision: perform all analyses **within each dataset** (intra-dataset).



$\Delta\beta$ correlation between GSE69914 and GSE225845.
Intersection CpGs show minimal concordance across platforms.



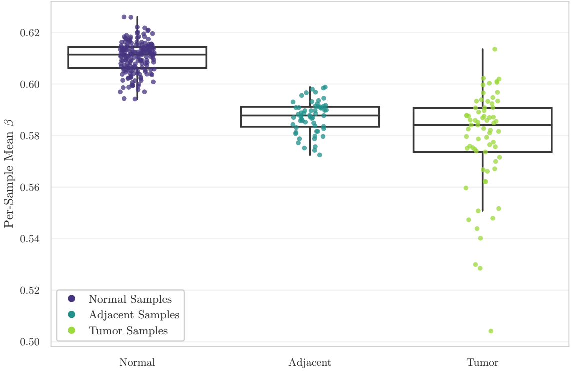
$\Delta\beta$ correlation between GSE287331 and GSE225845.
Cross-dataset signal remains inconsistent, confirming strong batch effects.

Table of Contents

3 Exploratory Data Analysis (EDA)

Global DNA Methylation Distributions

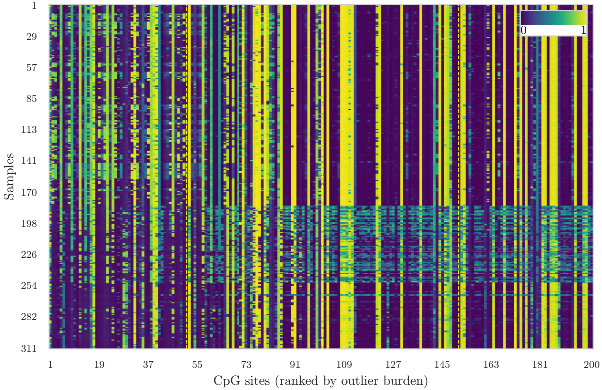
- Canonical bimodal β**-value distribution** (unmethylated vs. highly methylated CpGs).
- **Tumor samples**: shift towards **hypomethylation** and increased heterogeneity.
- **Adjacent samples**: early, subtle deviations from the Normal baseline.



Per-sample mean β-value distributions across tissue groups.
Tumor samples show global hypomethylation and higher variability compared to Normal and Adjacent samples.

Variability and Epigenetic Outliers

- **Unstable CpGs** are identified as those showing the highest number of extreme methylation values across samples.
- These CpGs highlight regions showing **focal hyper- or hypomethylation**.
- Local methylation instability is a hallmark of **biologically relevant epigenetic alterations**.

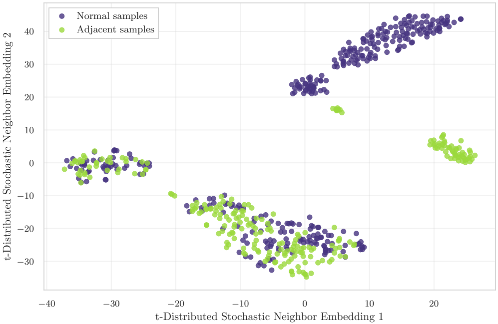


Heatmap of the 200 most unstable CpGs.

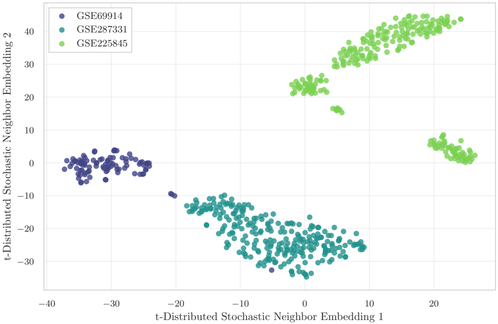
Colours represent β-values (blue = low methylation, yellow = high methylation).

Dataset Embeddings and Group Structure

- **GSE69914**: strong overlap between tissue groups.
- **GSE225845**: more complex and diffuse embedding structure.
- **GSE287331**: clear separation between **Normal** and **Adjacent**.



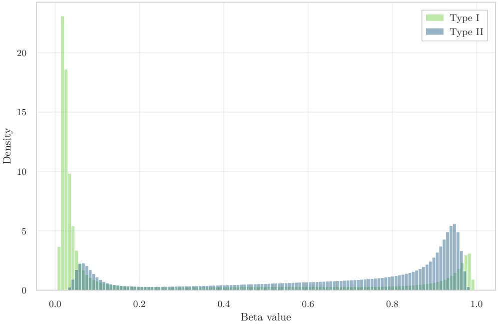
t-SNE embedding colored by tissue state.
Shows differences between Normal and Adjacent samples.



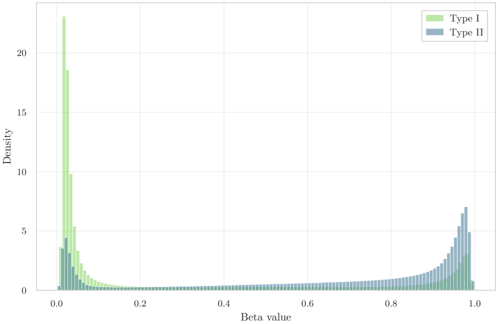
t-SNE embedding colored by dataset.
Highlights strong dataset-specific clustering and batch effects.

Probe-Design Bias and BMIQ Normalization

- Two probe designs (**Type I** vs. **Type II**) with incompatible β-value distributions.
- **GSE287331** shows a strong uncorrected probe-design bias.
- **BMIQ** performs quantile mapping of Type II probes toward the Type I distribution.
- Caution: BMIQ may reduce biological variability and partially mask real signals.



Pre-BMIQ: Type I vs. Type II distributions.
Strong mismatch between probe types highlights probe-design bias.



Post-BMIQ: Type I vs. Type II distributions.
BMIQ aligns Type II probes to Type I, but may dampen biological signal.

Table of Contents

Feature Selection in High Dimensions

- $p \gg n \rightarrow$ **dimensionality reduction** and filtering are essential.
- Technical filters: cross-reactive probes, SNP probes, non-CpG probes, Illumina manifest.
- Evaluate and correct potential batch effects (e.g., **COMBAT**).
- Three families of feature selection strategies:
  - **Statistical**: $\Delta\beta$, variance filtering, stability measures.
  - **Biological**: promoters, CpG islands, curated gene sets.
  - **ML-based**: Lasso, decision trees, embedded model selection.

- **Train / validation / test** with extremely small n (tens of samples).
- Generalising across datasets (A $\rightarrow$ B) is often **unreliable** due to batch and platform effects.
- **How many features** to select? **Which metrics** are stable in the $p \gg n$ regime?
- **Overfitting**, noise, and limited interpretability of complex models.
- Deep Learning: training from scratch is **infeasible** $\rightarrow$ only **pre-trained / transfer learning** approaches are realistic.

Approach	Data need	Interpretability	Robustness (small n)	Typical use-case here
Statistical	low–medium	high	high	$\Delta\beta$, variance, QC summaries
ML	medium–high	medium	medium–low	classification, feature selection
DL	very high	low	low	transfer learning / pre-trained models only

Comparison of modelling approaches under small-n / large-p constraints. Summary of data requirements, interpretability, robustness, and typical use-cases in this thesis.

Table of Contents
5 Conclusions

Take-home Message

- **Real-world bioinformatics pipeline**: from noisy GEO data and incomplete metadata to a **clean, analysable** methylation matrix via file conversion, manifest alignment, probe filtering, and batch assessment.
- **Epigenetics is deeply quantitative**: hundreds of thousands of CpGs, $p \gg n$, non-Gaussian β-distributions, strong batch and probe-design effects.
- **Role of mathematics and statistics**: exploratory analysis, stability and variability measures, dimensionality reduction, feature selection, and careful model evaluation under small-n/large-p constraints.
- **Machine learning in non-ideal conditions**: limited data, strong regularisation, preference for simpler and interpretable models; deep learning only via **transfer learning** or as a complement, not as a starting point.

Big picture: working at the interface of **biology, statistics, and computing** is challenging but extremely rewarding — and research projects like this are much closer to our skill set than they might appear.

Thank you!

If you have any questions, I'd be happy to answer.